

Characterisation of Neuronal Drebrin Expression in Serotonin Transporter Knockout Rats After MDMA Self-Administration

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Abstract

The aim of this study was to optimise a protocol to examine neuronal drebrin expression in the adult rat brain. A secondary aim was to assess drebrin concentrations in MDMA-untreated and MDMA-treated serotonin transporter (SERT) knockout (KO) and wildtype (WT) rats. This is because converging evidence implicates that SERT inhibition via these two mechanisms (genetic deletion or MDMA exposure) increases drebrin expression, but this drebrin-SERT relationship has not yet been investigated. After trialling both enzyme-conjugated immunohistochemistry and immunofluorescence (IF), a protocol using IF was optimised. An attempt at characterising drebrin expression in MDMA-treated and MDMA-untreated SERT KOs and WT rats was made using confocal microscopy. However, this proved unsuccessful at a quantitative level. Qualitative analysis showed that drebrin expression within each condition was inconsistent across biological repeats. Therefore, no conclusions could be made on differential drebrin expression as a result of SERT inhibition due to limitations in the current methodology. Future studies should employ more sensitive and efficacious approaches such as western immunoblotting and electron microscopy that will allow for drebrin quantification, to elucidate the role of SERT inhibition on cortical drebrin concentrations.

Introduction

The pharmacodynamics of common drugs of abuse, such as cocaine and amphetamine, primarily target the dopaminergic system by blocking and reversing the dopamine (DA) transporter (DAT), thereby enhancing DA neurotransmission (Gould et al., 2011; Harper, Langen & Schenk, 2014). All of these drugs of abuse, in turn, act on the dopaminergic pathway between the ventral tegmental area (VTA) and nucleus accumbens (NAcc), leading to stimulation of D₁ and D₂ receptors in the NAcc shell (Di Chiara, 2002). This dopaminergic activity occurs regardless of an individual drug's pharmacodynamic profile. Although drugs such as cocaine and amphetamine primarily act on the dopaminergic system, these drugs also act on the serotonin (5-HT) system albeit with lower affinities.

In contrast to cocaine and amphetamine, 3,4-methylenedioxymethamphetamine (MDMA), or 'ecstasy', is unique because it has the opposite pharmacodynamic profile. MDMA has greater affinity for the 5-HT transporter (SERT), by blocking its function and initiating reverse transport, at a level ten times greater than its affinity for the DAT (Gould et al., 2011). Compared to amphetamine, MDMA stimulates the serotonergic system 24 times more potently (Harper et al., 2014). Therefore, because MDMA primarily initiates 5-HT neurotransmission, it is not considered to possess the rewarding properties inherent to other drugs such as cocaine (Gould et al., 2011). As a result, there are two differing views on MDMA, with some believing it is a drug of abuse, while others posit that it lacks abuse liability because it does not produce the same effects as traditional drugs of abuse (Schenk, 2011). Because of these contrasting views, it is important to discern the associated neural changes occurring after MDMA exposure to further delineate the drug's abuse liability, particularly because MDMA is widely used among young adults (Schenk, 2011; Rozas et al., 2012; Ball, Wellman, Fortenberry & Rebec, 2009).

Multiple effects of MDMA exposure on both brain and behaviour have been demonstrated. At the behavioural level, acute MDMA has euphoric effects, but also produces feelings of depersonalisation and perceptual alterations (Abad et al., 2014; Rozas et al., 2012; Schenk, 2011). After chronic exposure, learning and memory deficits also arise (Abad et al., 2014), alongside increases in anxiety (Morini, Mlinar, Baccini & Corradetti, 2011). Changes underlying these chronic effects include the inhibition of cerebral 5-HT activity, the loss of 5-HT axon terminals, and a reduction in SERT expression (Gould et al., 2011; Morini et al., 2011). Human studies using positron emission tomography (PET) show a reduction in SERT binding among long-term MDMA users compared to healthy controls (Semple, Ebmeier, Glabus, O'Carroll & Johnstone, 1999). Similar PET results in rhesus monkeys after MDMA self-administration show reduced SERT concentrations in cortical areas such as the parietal lobe and cingulate cortex (Gould et al., 2011). As well, a binge regimen of MDMA in rats reduced 5-HT reuptake and overall 5-HT expression (Huff, Bhide, Schroering, Yamamoto & Gudelsky, 2013). Therefore, the neurotoxic effects of MDMA on the serotonergic system seem to be robust across species.

However, a number of factors determine how MDMA will affect the 5-HT system. For example, the SERT knockout (KO) rodent model has demonstrated that SERT deletion will alter how MDMA interacts with the serotonergic system. This model is generated by N-ethyl-N-nitrosurea (ENU) mutagenesis, resulting in a premature stop codon on exon 3 of the *Slc6A4* gene, which codes for the SERT (Homberg et al., 2007). This genetic targeting will result in offspring homozygous (HOM; SERT^{-/-}) or heterozygous (HET; SERT^{+/-}) for the SERT, and unaffected wildtypes (WT). In SERT^{-/-} animals, no SERT levels are detected in the brain, resulting in a nine-fold increase in extracellular 5-HT (Fabre et al., 2000), because there is no functioning reuptake transporter to decrease synaptic 5-HT activity (Homberg et al., 2007). In contrast, SERT^{+/-} animals will express only half the normal SERT levels observed in WTs.

This SERT KO model is intended to parallel the effects of a polymorphism found on the promoter region (5-HTTLPR) of the SERT gene in humans (Homberg et al., 2007). Those carrying the long 5-HTTLPR allele have normal SERT levels, while carriers of the short 5-HTTLPR allele have reduced SERT mRNA expression (Little et al., 1998). Short allele carriers also have reduced transcriptional SERT efficiency and decreased 5-HT reuptake activity (Lesch et al., 1996). Indeed, SERT^{+/-} offspring in the genetic knockout model mimic those humans who express the short allele for this polymorphism. Therefore, both the knockout model and the human short allele polymorphism result in the impaired functionality of the SERT, which dysregulates the serotonergic system and associated neural pathways.

Oakly, Brox, Schenk & Ellenbroek (2014) used this genetic KO model to determine whether acquisition rates for MDMA self-administration differed between SERT KOs and WTs, and whether SERT expression mediated the abuse liability of MDMA. All SERT KOs reached MDMA self-administration within 25 days, while only half of the WTs did so within this timeframe. SERT KOs also responded more during test sessions and had a higher breakpoint after introducing a progressive ratio, compared to WTs. These findings indicate that a genetic deletion of the SERT potentiates the reinforcing effects of MDMA, presumably because in the absence of the SERT, MDMA can only act on the DAT. This may result in greater MDMA-induced DA neurotransmission over 5-HT neurotransmission, in a way similar to other drugs of abuse such as amphetamine and cocaine. The association between SERT KOs and a potentiation of the rewarding properties of MDMA are consistent with human studies showing a relationship between the 5-HTTLPR polymorphism and drugs of abuse, particularly for MDMA (Cao, Hudziak & Li, 2013). Further, the finding that only half the WTs reached self-administration aligns with the hypothesis that an increase in synaptic 5-HT, at a greater level than DA, is not inherently reinforcing and protects against the rewarding properties of psychostimulants, in turn reducing a drug's abuse liability (Oakly et al., 2014). These findings

therefore indicate that the abuse liability of MDMA is determined by the relative concentrations of extracellular 5-HT and DA. As a result, greater 5-HT activity after MDMA will decrease its abuse liability, while greater DA activity after MDMA, via reduced SERT expression, will engender greater reinforcing qualities.

Nonetheless, the increases in extracellular 5-HT after genetic SERT deletion, or exposure to drugs such as MDMA, may also be initiating other changes at the molecular level. For example, structural brain changes at the level of dendritic spines have been observed after MDMA exposure. MDMA treatment increases spine density in NAcc and pyramidal medial prefrontal cortex (mPFC) neurons (Ball et al., 2009), as well as in MDMA-treated rats trained in the Morris water maze (Abad et al., 2014). These results are consistent with the increased spine density identified after administration of other drugs of abuse such as amphetamine and cocaine (Ball et al., 2009), indicating that morphological changes at the level of dendritic spines are a functional consequence of psychostimulant drug exposure. However, it is yet unclear what mechanisms underlie these increases in dendritic spine formation. The identification of these mechanisms will further discern the dynamic molecular changes occurring after drug exposure and could aid in the development of drug targets for the treatment of substance use disorders.

Changes in dendritic spine density and morphology are dependent on the actin cytoskeleton (Takahashi, Yamazaki, Hanamura, Sekino & Shirao, 2009). Of all the spine-related proteins, drebrin is one of the most highly localised proteins in spines that is associated with the actin cytoskeleton (Hayashi et al., 1996; Takahashi et al., 2009). Coded by the *DBN1* gene in humans, drebrin is an actin-binding protein that forms actin filaments (F-actin) and stabilises F-actin in dendritic spines (Koganezawa, Hanamura, Sekino & Shirao, 2017; Jung et al., 2015). Recent research from our lab has shown that drebrin is upregulated after genetic SERT deletion and MDMA self-administration. Using RNA sequencing in frontal cortex

tissue, drebrin mRNA expression in untreated SERT KOs was upregulated compared to untreated WT. This upregulation was increased even further for both KOs and WT that self-administered MDMA. The findings from our lab were novel and the relationship between drebrin, MDMA, and the 5-HT system has yet to be fully explored.

A limitation from our research is that no isoform-specific analysis on drebrin upregulation was conducted. In rodents and humans, drebrin has two isoforms produced by alternative splicing of the same gene, an embryonic type (drebrin E) and an adult type (drebrin A; Song et al., 2008; Kojima et al., 2016). Drebrin A contains the same mRNA sequence as the E type with a 46 amino acid sequence insertion in the middle of the mRNA strand (Ins2; Kojima et al., 2016). While drebrin E is ubiquitous and found in neuronal and non-neuronal tissue, drebrin A is a neuron-specific isoform found primarily in dendritic spines of cerebral synapses (Song et al., 2008).

The upregulation of drebrin mRNA expression in our lab is likely to be limited to the drebrin A isoform at dendritic spines. This is because the relative expression of drebrin E and drebrin A in neurons is dependent on neuronal development. Table 1 summarises the literature investigating the relative expression of drebrin E and A in the brain. These studies indicate that a drebrin isoform conversion occurs during development. As a function of neuronal maturation, drebrin A will increase, while drebrin E will decrease. Drebrin E will remain in the brain at low levels, but drebrin A expression will dominate, particularly in the cerebrum (Hayashi, Suzuki & Shirao, 1998; Aoki et al., 2005). The drebrin A-specific upregulation of drebrin after MDMA exposure is therefore more likely considering the WT and SERT KOs used in our RNA sequencing study had reached adulthood.

Drebrin A upregulation may be increasing dendritic spine density via its role in synaptic communication. Because the time course of drebrin A expression is similar to that of synaptophysin and occurs alongside synapse formation, the gradual upregulation of drebrin A

during development has been implicated in synaptogenesis (Hanamura, Kamata, Yamazaki, Kojima & Shirao, 2018; Hayashi et al., 1996; Hayashi et al., 1998). Drebrin A appears during the primary stages of synaptogenesis (Aoki et al., 2005), located postsynaptically in cortical neurons. Indeed, the clustering of drebrin A in postsynaptic dendritic spines will occur before the formation of the postsynaptic density (PSD) and the accumulation of presynaptic vesicles, indicating its crucial role in the establishment of synaptic communication (Aoki et al., 2005). Drebrin A clustering is also dependent on the type of synapse. Greater drebrin concentrations are found in asymmetric synapses where excitatory glutamatergic signalling occurs (Kobayashi et al., 2004; Takahashi et al., 2009; Aoki et al., 2005).

Table 1.

Summary of findings on the relative expression of drebrin E and drebrin A. *Note.* PND = postnatal day; RT-PCR = reverse-transcriptase polymerase chain reaction; DIV = days *in vitro*.

Method	Drebrin E Expression	Drebrin A Expression	Additional Notes	Reference
Western Immunoblotting	Detectable at PND0	Detectable at PND10, gradual increase through to PND21	Drebrin E detectable in both cerebellar and cerebral cortices; drebrin A expression high in cerebral cortex but low in cerebellar cortex	Aoki et al. (2005)
	Decrease occurring at PND28	Gradual upregulation begins at PND7	Drebrin A expression in cerebellum remained lower than drebrin E expression until at least PND70	Hayashi et al. (1996)
RT-PCR (<i>in vivo</i>)	mRNA intense at PND7; mRNA faint at PND14	mRNA faint at PND7; mRNA intense at PND14	-	Hayashi et al. (1998)
RT-PCR (<i>in vitro</i>)	mRNA intense at 8 DIV; mRNA faint at 14 DIV	mRNA faint at 8 DIV; mRNA intense at 14 DIV	Used cortical neurons dissociated from a 20-day old foetal rat brain	Hayashi et al. (1998)

As stated, the role of drebrin A in these spines is to bind to and stabilise F-actin. Because the cytoskeletal structure in dendritic spines requires stable F-actin, drebrin mediates dendritic spine morphology (Hanamura et al., 2018). Hanamura et al. (2018) showed that the drebrin E to A conversion is required for F-actin stabilisation, such that the location of the Ins2 sequence in the middle of the drebrin A mRNA strand produces its ability to stabilise F-actin and maintain spine morphology. Because spine morphology and synaptic plasticity underlie learning and memory, these results indicate that drebrin A has a role in these higher-order functions (Jung et al., 2015).

Drebrin A KO (DAKO) and knockdown models in mice have demonstrated the functional consequences of an absence of drebrin A in spines. Drebrin A knockdown increased spontaneous locomotor activity in an open field test and produced perseveration of inappropriate responses in the Morris water maze, indicating learning-based deficits (Kobayashi et al., 2004). DAKO mice also have impaired fear learning, shown in tone-shock fear conditioning paradigms (Kojima et al., 2016). A downregulation of drebrin A in humans has also been implicated in mild cognitive impairment, Alzheimer's disease and Down syndrome (Counts et al., 2012; Jung et al., 2015). Therefore, a lack of drebrin A in dendritic spines will result in widespread learning, memory and cognition deficits in rodents and humans, due to the destabilisation of dendritic spines, and therefore the reduced efficacy of synaptic plasticity.

While these learning and memory deficits of a drebrin A downregulation are clear, the association between drebrin and MDMA is less clear. It is surprising that there have been no investigations on the relationship between drebrin and drug addiction, given that adaptations in dendritic spine densities have been reported both in learning and memory (Uhernik, Montoya & Smith, 2018), and after drug exposure (Ball et al., 2009). Nonetheless, there is evidence for the potential relationship between drebrin and 5-HT, other than data from our lab, to suggest

that drebrin could also be associated with MDMA. Roppongi, Kojima, Hanamura, Yamazaki and Shirao (2013) showed that fifteen minutes of exposure to a 5-HT_{2A} agonist in hippocampal neurons *in vitro* reduced both drebrin and F-actin clustering in dendritic spines but had no effect on spine morphology nor on other synaptic markers such as synaptophysin and PSD-95. This indicates that 5-HT_{2A} activation has a specific effect on drebrin and F-actin, instead of a general effect that initiates the loss of synapses.

However, Yang et al. (2003) showed the opposite effect of serotonergic activity on drebrin expression. Hippocampal drebrin A mRNA expression was upregulated in chronically-stressed rats treated with fluoxetine compared to untreated controls. Because fluoxetine is a selective serotonin reuptake inhibitor (SSRI), it may be that an increase in 5-HT activity via blocking of the SERT enhances drebrin expression, whereas an increase in 5-HT activity via 5-HT_{2A} agonism reduces drebrin expression as Roppongi et al. (2013) demonstrate. Although the latter was an *in vitro* study and Yang et al. was *in vivo*, these contrasting findings of 5-HT activation on drebrin highlight that 5-HT mechanisms altering drebrin concentrations are dynamic and complex. This is further complicated by there being at least 14 5-HT receptor subtypes, with 5-HT interacting with these subtypes at different sensitivities (Donaldson, Nautiyal, Ahmari & Hen, 2013). Therefore, 5-HT activity could be producing differential effects on these receptors and to postsynaptic markers such as drebrin, compared to the activation of a single receptor subtype.

Alongside findings from our lab, the data by Yang et al. (2003) sheds light on how MDMA exposure could influence drebrin concentrations in dendritic spines. Because both MDMA and fluoxetine block the SERT, MDMA exposure may also result in an upregulation in drebrin expression. This hypothesis is consistent with studies showing that MDMA increases dendritic spine density (Ball et al., 2009; Abad et al., 2014), which could be mediated by the upregulation of drebrin A. Similar increases in drebrin A expression may also be found in

SERT KO rats, because it increases 5-HT levels via reduced SERT expression akin to MDMA and fluoxetine.

Because the analysis of drebrin in our lab has been limited to mRNA expression in the frontal cortex, a brain-wide examination of drebrin is warranted. Therefore, the aim of this study was to optimise a protocol for examining drebrin expression, and to characterise this expression in WTs and SERT KOs exposed to MDMA self-administration or unexposed to MDMA. Drebrin expression in sagittal rat brain sections was characterised using immunohistochemistry (IHC) with an antibody against drebrin detecting both isoforms (Shirao & Obata, 1986). It was hypothesised that SERT inhibition, via genetic targeting or drug exposure, would upregulate drebrin expression (Yang et al., 2003; Ball et al., 2009; Abad et al., 2014). If this is the case, both MDMA exposure and the SERT KO model would enhance drebrin concentrations, compared to WTs and those not exposed to MDMA.

Method

Animals

MDMA-Unexposed Animals

The brains of 6 adult Wistar rats were used, of which 3 were homozygous (SERT^{-/-}) KOs and 3 were WT (SERT^{+/+}), bred using a HET (SERT^{+/-}) x HET (SERT^{+/-}) pairing. After PND20, each rat was socially isolated for four weeks. During this period social preference was measured using conditioned place preference. They were exposed to ketamine for the surgical implantation of a heart rate variability device, as well as carprofen and penicillin during the post-surgery period. During adulthood, rats were exposed to mild stress via saline injection or exposure to a novel environment without food or bedding. Two days after mild stress induction, when subjects had reached at least PND80, they were consciously decapitated. Brains were extracted and snap frozen in liquid nitrogen within 40 seconds of decapitation and were subsequently stored at -80°C until tissue preparation. No controls unexposed to these conditions were available.

MDMA-Treated Animals

The brains of 6 male adult Wistar rats were used, of which 3 were homozygous (SERT^{-/-}) KOs and 3 were WT (SERT^{+/+}). Before the onset of MDMA self-administration, animals were separated into individual cages. For the implantation of the self-administration catheter, subjects were exposed to ketamine, xylazine, and carprofen, as well as heparin and penicillin G potassium during the post-surgery period. Animals were then placed into self-administration chambers daily, under fixed ratio (FR) 1, FR2, and FR5 schedules. They then underwent a progressive ratio (PR) schedule, with the last PR session preceding a final FR5 schedule to resume normal responding. Each successful lever press resulted in an infusion of MDMA (BDG Synthesis, Wellington, NZ), prepared from powder mixed with heparinised saline, at a dose of between 0.25mg/kg to 1mg/kg. After the final self-administration session, subjects

were euthanised via CO₂ asphyxiation. Brains were extracted and frozen immediately at -80°C until tissue preparation.

Tissue Preparation and Sectioning

Brains were removed from -80°C storage and thawed in a -20°C freezer for 30 minutes. Brains were cut along the longitudinal fissure to separate the left and right hemispheres, which were then immersed into separate tubes containing 4% paraformaldehyde (PFA) in 1M phosphate buffer, pH 7.4 for fixation. Fixation occurred for 1 hour at 4°C. Brain hemispheres were removed from PFA and immediately placed into tubes containing 30% sucrose made in 1M phosphate buffer, pH 7.4 for cryoprotection at 4°C. They remained in the sucrose solution until they sank into the tubes, after which they were removed and dried with paper towels. Hemispheres were immediately snap frozen in isopentane (2-methylbutane) cooled with dry ice and were stored at -80°C until sectioning. Prior to sectioning, brain hemispheres were embedded in Tissue-Tek® optimum cutting temperature media (OCT; Global Science, Auckland, NZ) in a -20°C chamber. Sagittal sections were cut at 10µM thickness using a Leica CM3050 S Research Cryostat (Leica Biosystems, Wetzlar, Germany), and were mounted onto Superfrost® Plus slides (Interlab, Wellington, NZ). Slides were stored at -80°C until they were used for immunohistochemistry.

Enzyme-Conjugated Immunohistochemistry

Slides were removed from -80°C storage and brought to room temperature (RT) for 30 minutes. They were fixed once more in acetone for 15 minutes and washed by immersion in phosphate buffered saline (PBS) with 0.3% Triton X-100 (PBST) 3-4 times, after which they were air dried under a fume hood for 30 minutes. Slides were immersed in PBST for permeabilization at RT for 20 minutes with mild agitation. To reduce nonspecific binding, sections were blocked with 3% bovine serum albumin (BSA) in PBST for one hour at RT. They were then incubated with the primary antibody overnight at 4°C, diluted in 3% BSA in

PBST. The next day, slides were washed three times in PBST for five minutes each with mild agitation. They were incubated with a biotinylated anti-mouse IgG secondary antibody (1:1000) in PBST for 90 minutes at RT, and subsequently washed three times in PBST with mild agitation. Antibody localisation was performed using the avidin-biotin-peroxidase complex (Vectastain ABC Kit, Vector Laboratories, Burlingame, CA, USA) in PBST (1:1000) for one hour at RT. The slides were washed three times in PBST with mild agitation. The immunoreaction was visualised using 3,3'-diaminobenzidine (DAB). After air drying for 30-60 minutes, slides were immersed in increasing concentrations of ethanol (70%, 95%, 100%), cleared using Histo-Clear[®], mounted with DPX, and affixed with a coverslip.

Fluorescent Immunohistochemistry

Slides were removed from -80°C storage and brought to RT for 30 minutes. They were fixed in acetone for 15 minutes and washed in 1x Tris-buffered saline (TBS) with Tween-20 (TBST) three times for 5 minutes each. To reduce nonspecific binding, slides were blocked with 5% BSA in 1x TBST, which contained 0.3M glycine to diminish background fluorescence. Slides were blocked for 60-120 minutes, then washed three times in TBST for five minutes each. Slices were incubated with the primary antibody, diluted in 1% BSA dissolved in TBST, overnight at 4°C in a darkened and humidified chamber. The next day, slides were washed three times in TBST for five minutes each. They were then incubated with the appropriate secondary antibody diluted in 1% BSA in TBST, in a darkened chamber at RT for 120 minutes. After washing the slides in TBST three times for a total of fifteen minutes, slides were mounted in ProLong Gold Antifade (Invitrogen, Thermofisher Scientific, Auckland, NZ) which contained cell nuclei counterstain 4',6-diamidino-2-phenylindole (DAPI; Invitrogen, Thermofisher Scientific, Auckland, NZ), and affixed with coverslips. Table 2 displays the concentration and source information of all primary and secondary antibodies used, including primary antibodies used as positive controls.

Table 2.

Primary and secondary antibodies used in this study with dilution, source, and usage information. *Note.* Enzyme = Enzyme-conjugated IHC; IF = Immunofluorescence. GFAP = glial fibrillary acidic protein.

Primary Antibodies				
Marker (clone)	Antibody Raised in	Antibody Dilution	Antibody Source	Usage
Drebrin (M2F6)	Mouse	1:100 (enzyme); 1:200, 1:1000 (IF)	MBL Life Science (D029-3); Abcam (ab12350)	Enzyme; IF
Parvalbumin (PARV-19)	Mouse	1:5000	Sigma-Aldrich (P3088)	Enzyme
GFAP	Rabbit	1:1000	Dako, Agilent (Z0334)	IF
Secondary Antibodies				
Secondary Antibody		Antibody Dilution	Antibody Source	Usage
Biotinylated anti-mouse IgG		1:1000	Vector Laboratories (BA-9200)	Enzyme
Rhodamine Red TM -X anti-mouse IgG		1:200	Invitrogen (R6393)	IF
Alexa Fluor 568 anti-mouse IgG		1:500	Abcam (ab175473)	IF
Alexa Fluor 488 anti-mouse IgG		1:500	Invitrogen (A11001)	IF
Alexa Fluor 555 anti-rabbit IgG		1:500	Invitrogen (A21429)	IF

Autofluorescence Quenching Methodology

Three different methods were used to eliminate autofluorescence, particularly of lipofuscin, a fluorescent pigment found in the cytoplasm of neurons (Schnell, Staines & Wessendorf, 1999). Sagittal sections of a rat brain not from the sample used in this study were treated with Sudan black, copper sulphate and sodium borohydride. Protocols for these methods were based on Schnell et al. (1999). After blocking with 5% BSA in TBST with

0.3M glycine for 60 minutes, separate tissue sections were treated with these three methods.

ProLong Gold Antifade with DAPI was applied to the tissue immediately after autofluorescence quenching, and the slides were affixed with a coverslip.

Sudan Black ($C_{29}H_{24}N_6$)

Sudan black (0.3% w/v) was prepared in 70% ethanol. The solution was mixed constantly for 60-120 minutes in a darkened chamber at RT, then pipetted directly on the tissue for 10 minutes. The tissue was subsequently washed in 1x TBS 8 times for 5 minutes each.

Copper Sulphate ($CuSO_4$)

$CuSO_4$ (10mM) was prepared in ammonium acetate buffer (50mM, CH_3COONH_4). The ammonium acetate buffer was brought to pH 5.0 using glacial acetic acid. $CuSO_4$ was pipetted onto the tissue twice at 10 minutes each, then washed in 1x TBS three times for a total of fifteen minutes.

Sodium Borohydride ($NaBH_4$)

A $NaBH_4$ solution was prepared in fresh 1x TBS at a concentration of 1mg/ml. Tissue was treated with $NaBH_4$ by pipetting the solution directly onto the tissue twice for ten minutes each. Slides were then washed in 1x TBS three times for a total of fifteen minutes.

Microscopy

Enzyme-conjugated and fluorescence immunostained tissue was examined using an Olympus IX58 inverted microscope (Olympus, Hamburg, Germany), with an Olympus TH4-200 halogen lamp (Olympus Corporation, Tokyo, Japan) used for enzyme-conjugated IHC tissue and an X-Cite® 120Q light source for IF (Excelitas Technologies, MA, USA). Sections for confocal microscopy were viewed using an Olympus Fluoview FV1000 confocal laser-scanning biological microscope (inverted model IX81) fitted with a heating chamber (INU-21LCS-F1, Tokai Hit Co., Ltd, Japan).

To enable accurate visualisation of the images in this thesis, some required artificial increase of the exposure as reimaging was not possible. Exposure was artificially increased on select figures using Preview (v9.0, MacOS, Apple Inc.) as stated in the figure captions.

Results

An initial attempt to examine neuronal drebrin expression was performed using enzyme-conjugated IHC compared to a negative control (Figure 1). Little differences were seen between drebrin-stained and negative control tissue. Although a colour change was evident in the tissue stained against drebrin, there could be a number of reasons why no specific staining was detected, such as the expired Vectastain ABC kit used or degradation of the drebrin antibody after multiple instances of freezing and thawing. As well, there was no positive control to validate the enzyme-conjugated IHC protocol.

A second attempt was made to characterise drebrin expression in neuronal tissue, which included parvalbumin (PV) as a positive control to validate the IHC protocol (Figure 2). Because the aim of this attempt was protocol validation, the brain tissue used was from an adult Wistar WT rat, not from the sample in the current study, that had been embedded in OCT. Yet again, there were no distinct differences between the primary antibodies and negative control. Staining for PV was unsuccessful with no identifiable PV interneurons. Nonetheless, the drebrin-stained tissue showed some level of positive staining compared to the negative control. Because the PV staining was unsuccessful, it was difficult to ascertain whether the drebrin-stained tissue depicts drebrin-positive cells or whether the observable staining was a product of nonspecific binding. Moreover, the issues with the first attempt, namely the degradation of the drebrin antibody and expired Vectastain ABC kit, were not addressed in this second attempt and could be causing these issues. Therefore, in the third attempt the aim was to use a new visualisation method, immunofluorescence (IF), to determine whether these issues persisted.

Figure 3 depicts the initial IF results using Rhodamine Red™-X as the secondary antibody. Drebrin-stained tissue was more fluorescent than the negative control in a diffuse and ubiquitous manner. While some staining was evident, a stronger signal-to-noise ratio was

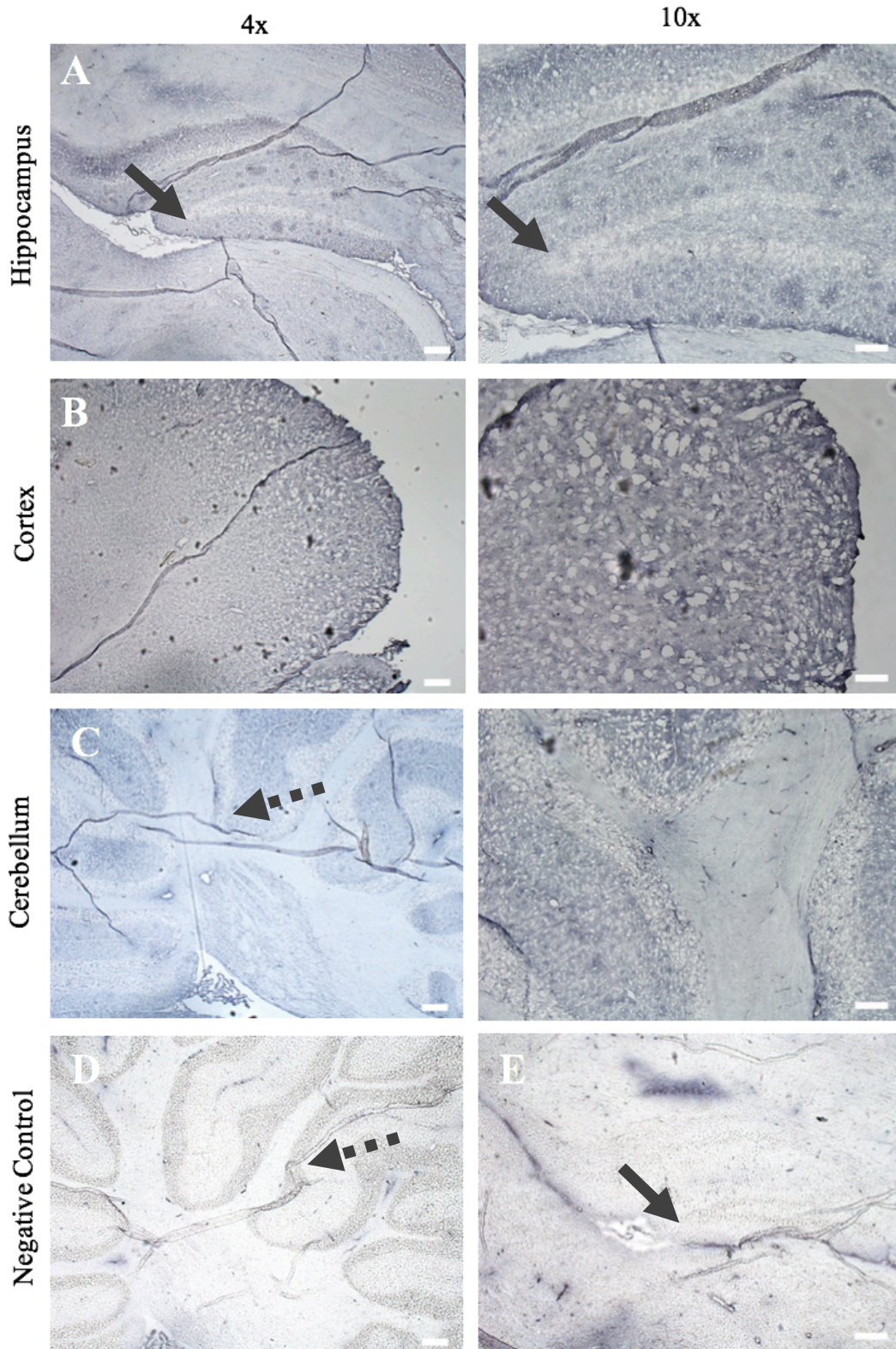


Figure 1. Initial drebrin staining using enzyme-conjugated IHC. Tissue displays hippocampal (A), cortical (B), and cerebellar (C) regions at 4x and 10x magnification. Negative control images depict the cerebellum (D) and hippocampus (E), both at 4x magnification. Dashed arrows show the cerebellar granular layer, and solid arrows show the dentate gyrus granular layer, with little differences between drebrin-labelled and negative control tissue. Scale bars = 200 μ m (4x magnification), 100 μ m (10x magnification).

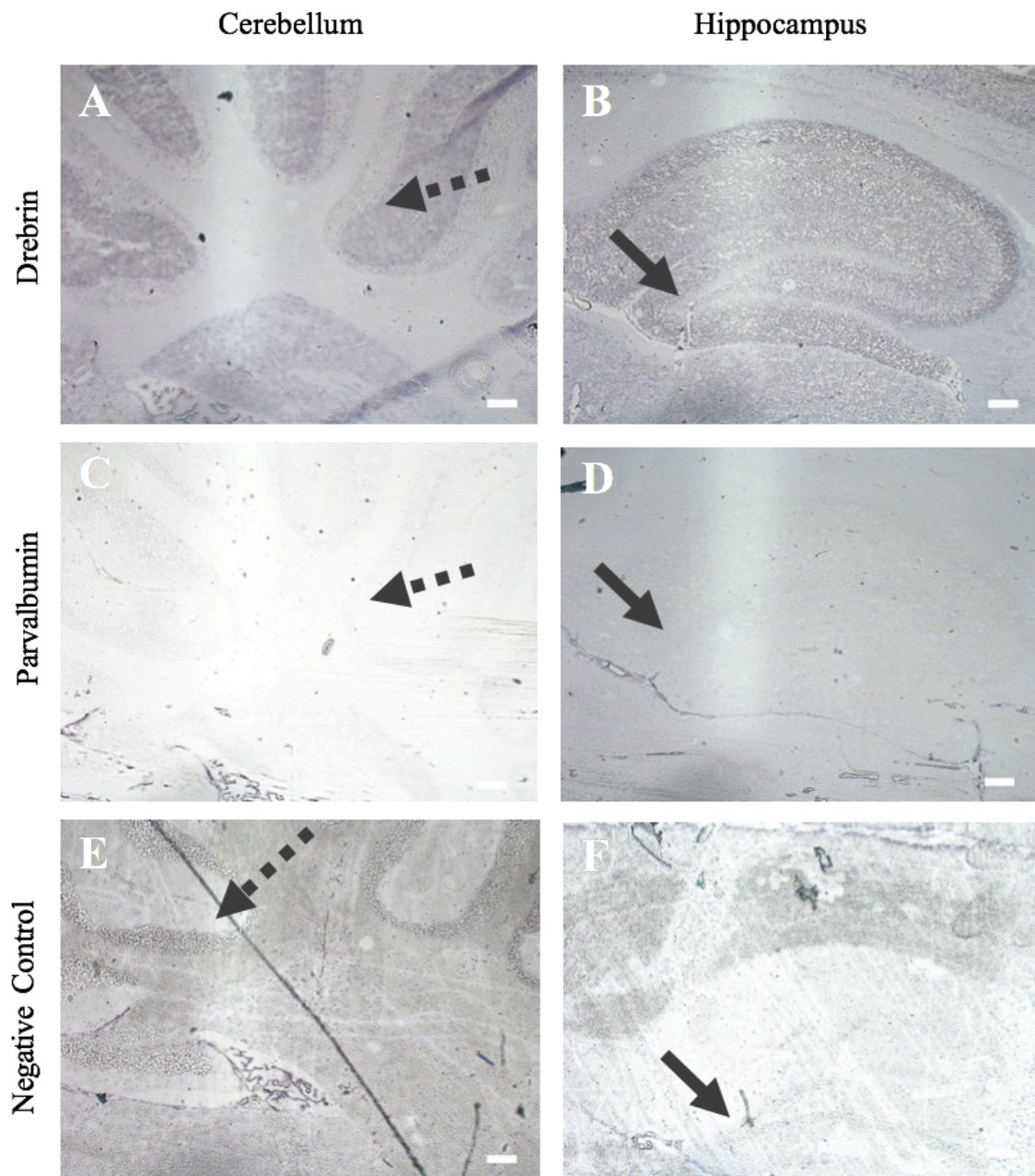


Figure 2. Follow-up enzyme-conjugated IHC results for troubleshooting purposes. Panels show drebrin staining in the cerebellum (A) and hippocampus (B). Parvalbumin was used as a positive control in these same regions (C, D), as well as a negative control (E, F). Dashed arrows indicate the cerebellar granular layer, while solid arrows show the dentate gyrus. Scale bar = 200µm.

required to more accurately assess differences between drebrin and negative control tissue. The high autofluorescence in the negative control tissue, especially in the cortex, may have lowered this signal-to-noise ratio. Therefore, reducing this autofluorescence was a priority.

The methodology from Schnell et al. (1999) was next employed to quench this autofluorescence. Unquenched negative control tissue was compared with copper sulphate, sodium borohydride, and Sudan black. Both sodium borohydride and copper sulphate eliminated autofluorescence, compared to the negative control (Figure 4). Quenching with Sudan black also eliminated the autofluorescence but this method was undesirable due to the intense black staining of the tissue after the quenching protocol. While both sodium borohydride and copper sulphate were effective quenching methods, the former method was employed in all subsequent IF protocols before blocking tissue with 5% BSA. The only change, however, was that NaBH₄ was now pipetted onto slides thrice for ten minutes each.

Next, an antibody against glial fibrillary acidic protein (GFAP) was used as a positive control on brain tissue to validate the IF protocol (Figure 5). Clear differences between GFAP-labelled and negative control tissue validated the IF protocol employed in this study.

Because the IF procedure was validated, Alexa Fluor 488 was used instead of Rhodamine RedTM-X to ensure that the diffuse staining was not due to an ineffective secondary antibody. A secondary aim of this next experiment was to examine the drebrin-stained tissue under higher magnifications using confocal microscopy. Because drebrin A should be expressed only in dendritic spines (Koganezawa et al., 2017), an examination of the tissue under high magnification may show more localised fluorescent labelling of drebrin. There remained ubiquitous staining using this alternative fluorophore at low magnifications, indicating that the initial secondary antibody (Rhodamine RedTM-X) was not ineffective. However, this diffuse expression was eliminated when tissue was examined at 100x magnification in the ventral cerebrum (Figure 6). Fluorescent labelling in the ventral cerebrum

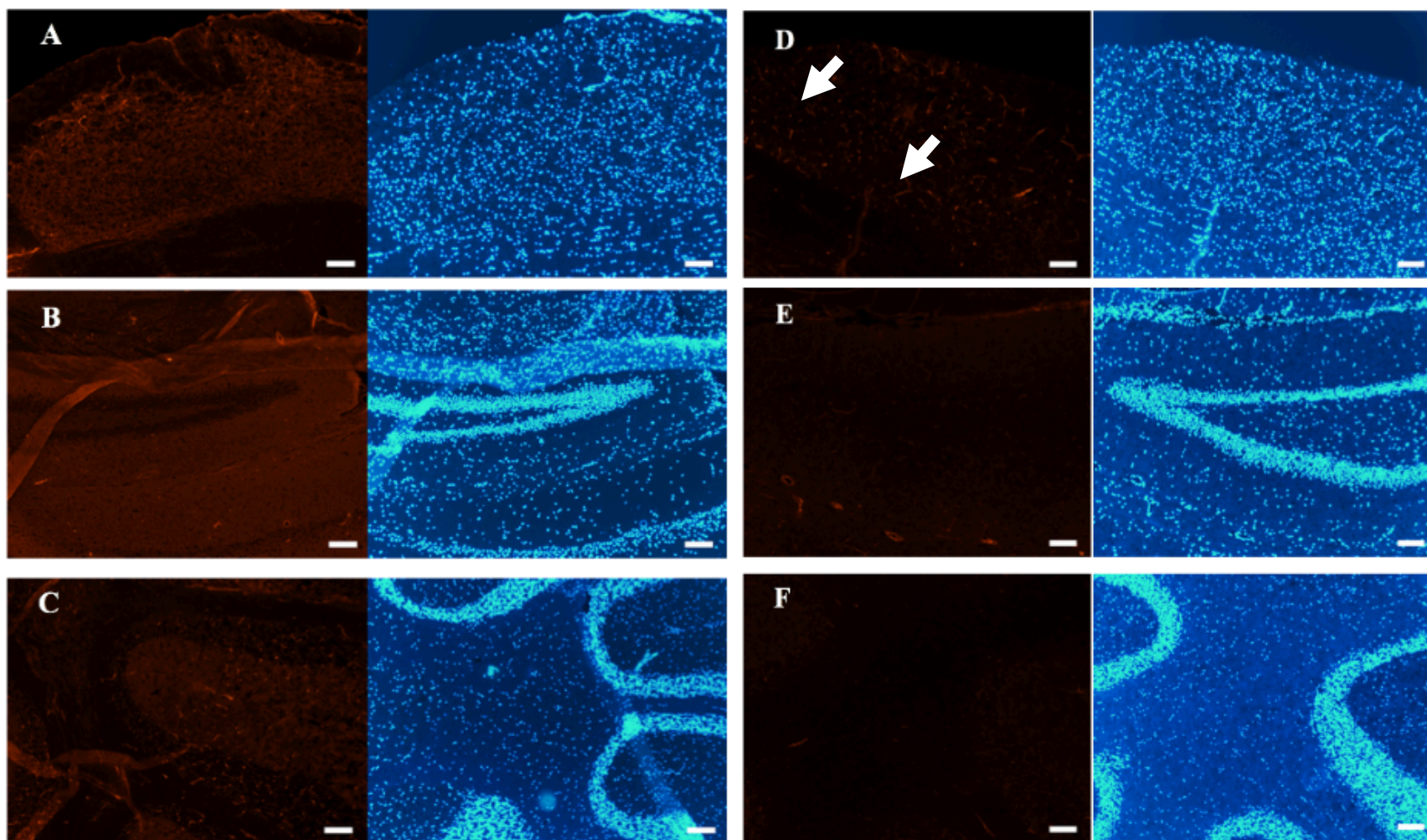


Figure 3. Initial IF results, depicting drebrin (A-C) and negative control (D-F) in red, alongside DAPI (blue). Images show frontal cortex (A, D), dentate gyrus (B, E), and cerebellum (C, F). Autofluorescence in the negative control tissue is denoted by the white arrows. Scale bar = 100 μ m.

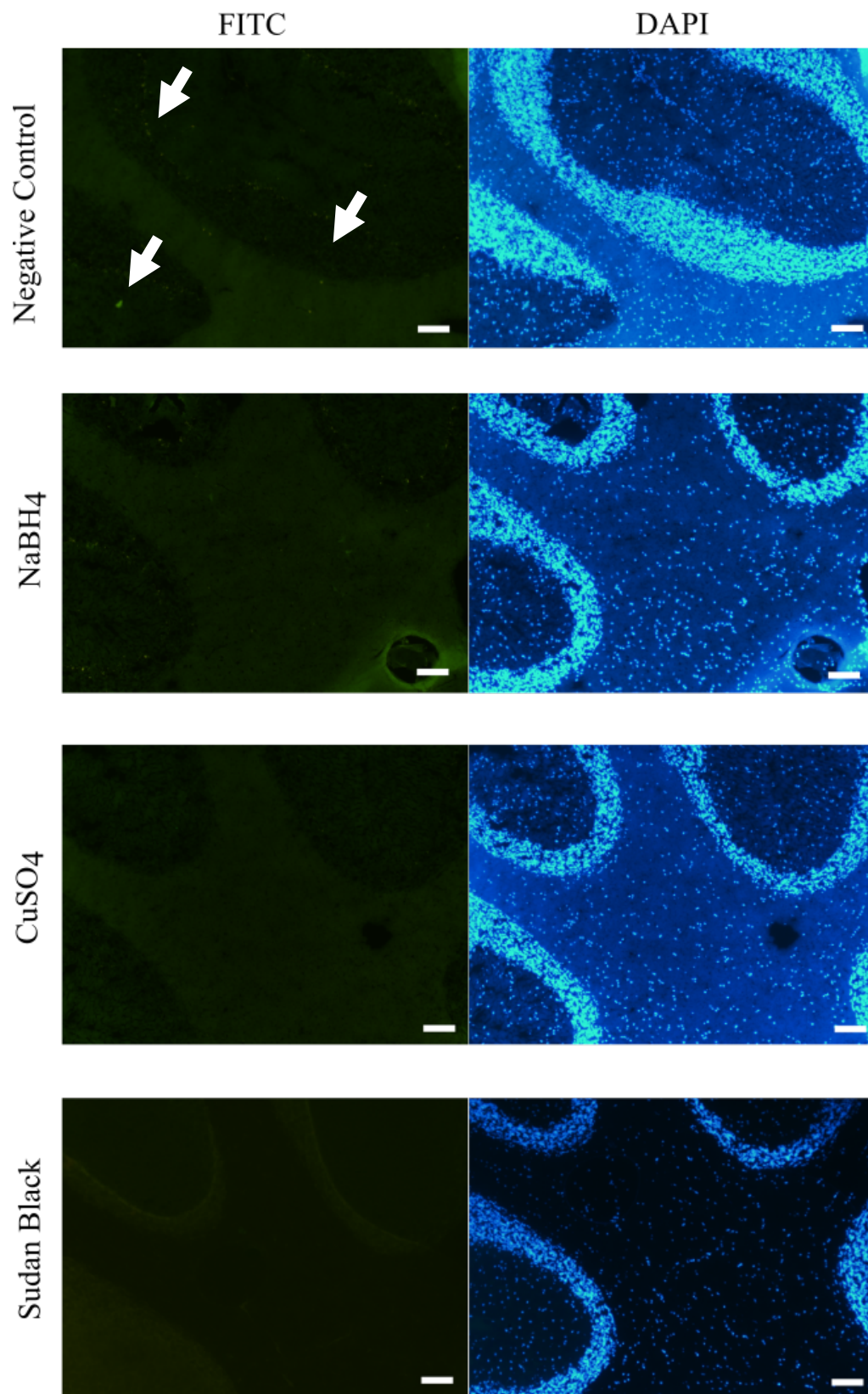


Figure 4. Autofluorescence quenching comparing sodium borohydride (NaBH_4), copper sulphate (CuSO_4), and Sudan black with a negative control using the FITC excitation filter alongside DAPI. White arrows indicate more intense autofluorescence in the negative control compared to tissue where autofluorescence was quenched. Scale bar = $100\mu\text{m}$.

was punctate and more identifiable at higher magnifications. The punctate labelling in the drebrin-labelled tissue is suggestive of dendritic spines where drebrin A should be expressed. Drebrin-labelled tissue with Rhodamine Red™-X as the secondary antibody was also examined at these higher magnifications (Figure 7), showing similar punctate characterisation as seen in Figure 6. Insofar as this expression is suggestive of dendritic spines, an IF protocol for examining drebrin expression was successfully optimised.

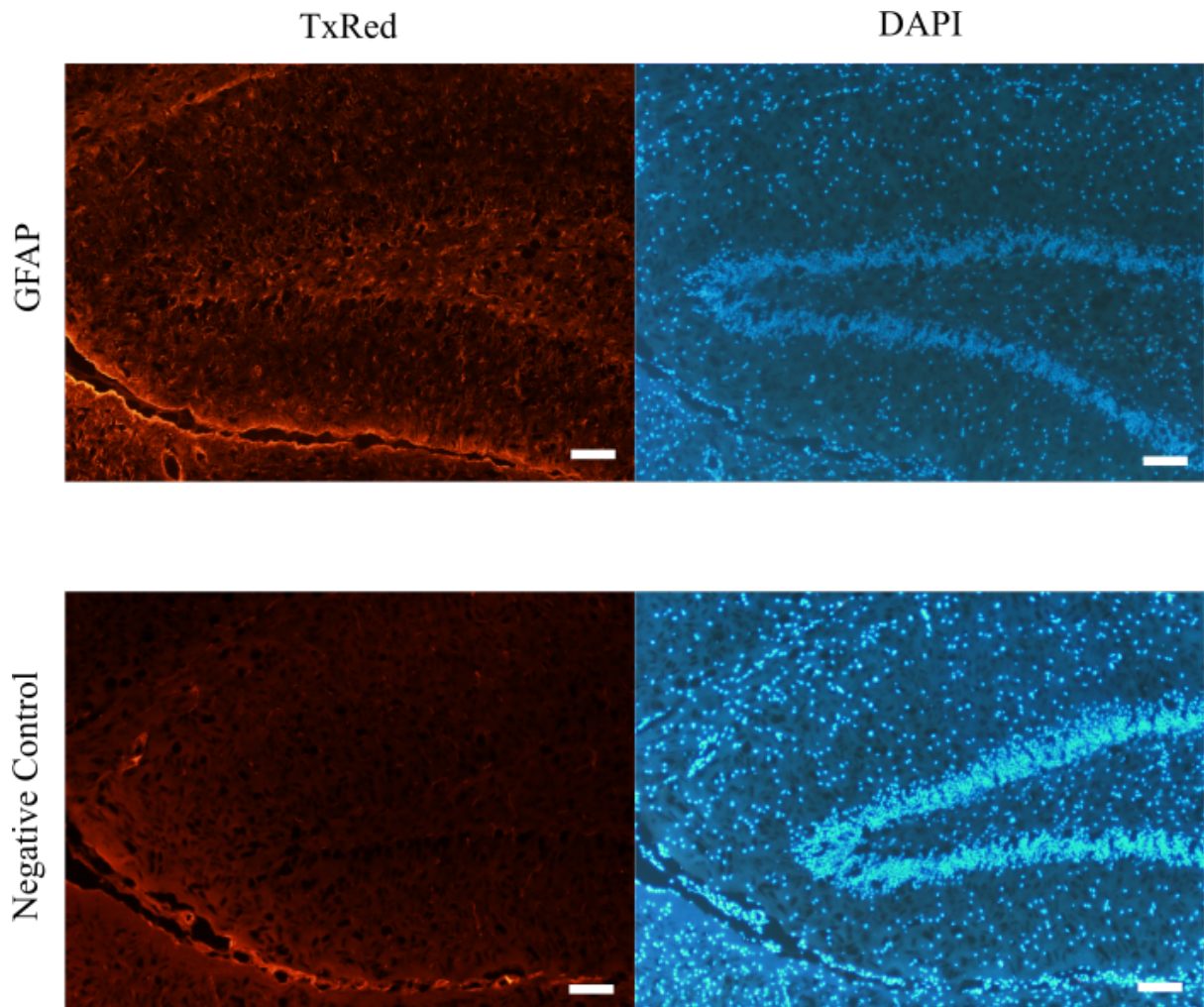


Figure 5. Dentate gyrus tissue stained against GFAP using the TxRed excitation filter with counterstain DAPI. The intense fluorescence in GFAP tissue in contrast to the low-intensity fluorescence in the negative control validates the IF protocol used here. Alexa Fluor 555 was the secondary antibody used for GFAP-labelled tissue. Scale bar = 100µm.

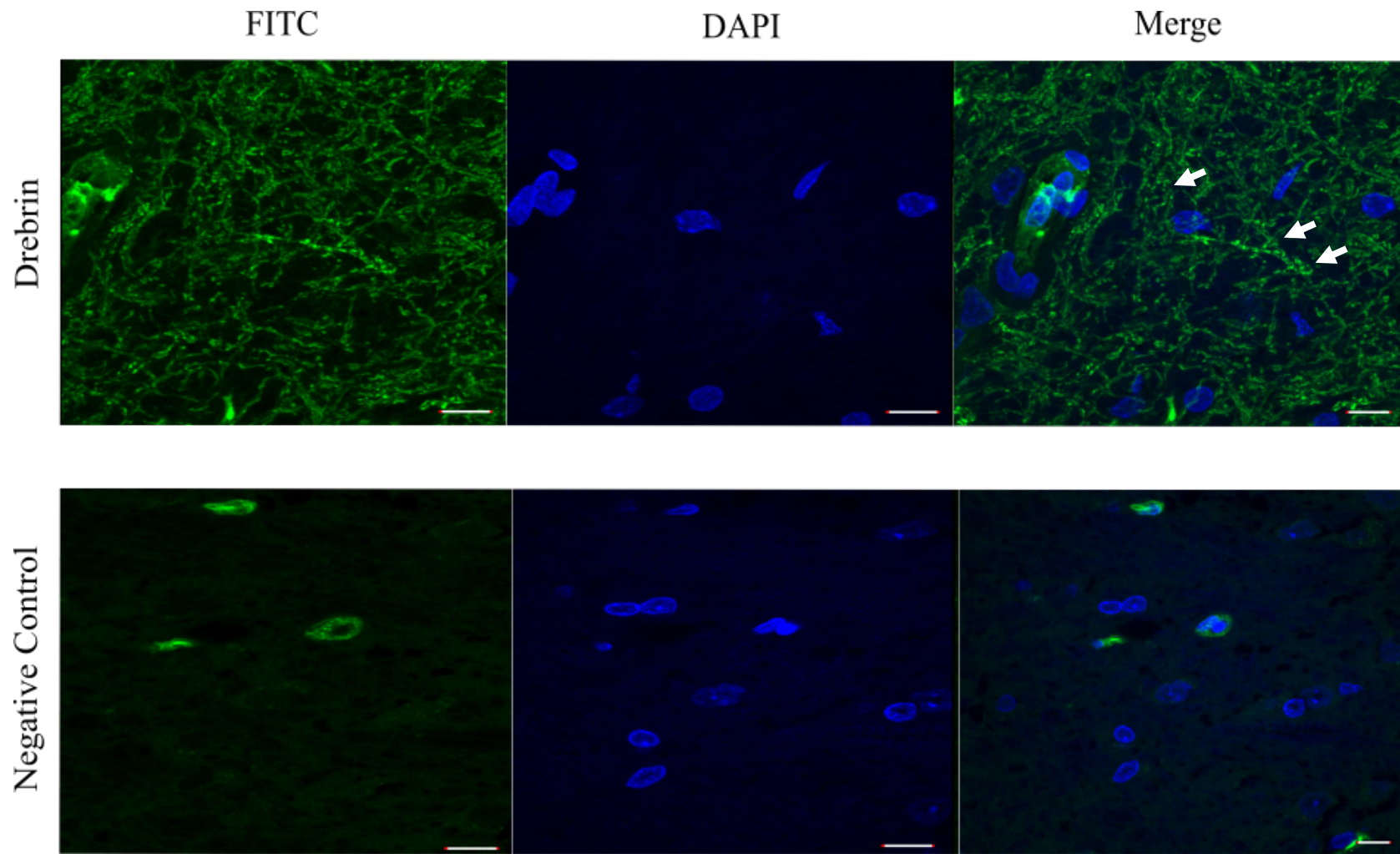


Figure 6. Drebrin fluorescence in the ventral cerebrum using confocal microscopy with Alexa Fluor 488, compared to a negative control. The punctate spots of fluorescence throughout the drebrin-labelled tissue are suggestive of drebrin A localisation in dendritic spines (white arrows). Scale bar = 5 μ m.

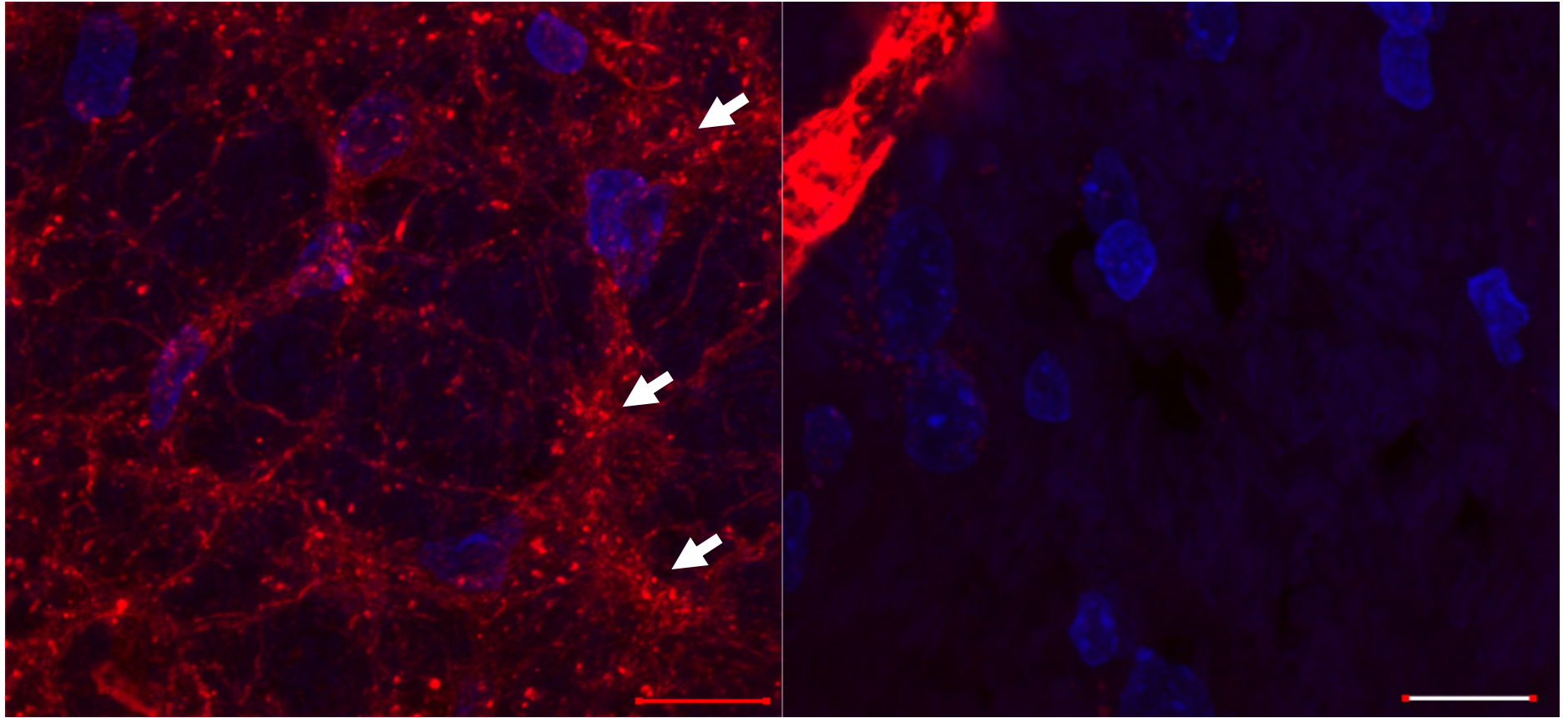


Figure 7. Drebrin fluorescence in the ventral cerebrum (left) using confocal microscopy with Rhodamine RedTM-X, compared to negative control (right), with cell nuclei counterstain, DAPI (blue). Similar to Figure 6, the punctate spots of fluorescence throughout the tissue corroborates the positive drebrin localisation in dendritic spines (white arrows). Scale bar = 10 μ m.

Drebrin expression was subsequently compared between the treatment x genotype conditions in three biological repeats (Figs. 8-10).¹ The images of drebrin-labelled tissue and negative controls are representative of overall staining in the frontal cortex. A qualitative comparison of drebrin expression was relied upon because it was not possible to quantitatively examine drebrin concentrations between the conditions. Results were inconsistent across the three repeats. In the first test (Figure 8), the brightest fluorescent intensity in MDMA-treated WT tissue compared to the other three groups suggested drebrin expression was highest in this condition. However, repeat staining did not demonstrate a similar level of fluorescence in this condition (Figs. 9-10). These inconsistencies across repeats were also evident in the remaining three conditions, such that there was low fluorescence in one repeat but high fluorescence in others. These contradictions suggest that visual observation alone was an unreliable way to determine whether drebrin expression was differentially expressed between these conditions. Hence, a subtler method to examine differential drebrin expression was required.

Importantly, however, across all tests no expression was found either in the negative control tissue nor in the cerebellum of drebrin-labelled tissue. Consistent with previous research (Hayashi et al., 1996), the lack of expression in the cerebellum corroborates that the protocol for drebrin A localisation in dendritic spines was optimised, even though a quantitative analysis of drebrin expression across conditions was not possible using the current methodology.

¹ Alexa Fluor 568 (1:500) was the secondary antibody used in these IF tests. Further, after a test comparing different concentrations of the M2F6 drebrin antibody (1:200, 1:500, 1:1000), the concentration in these tests was lowered to 1:1000.

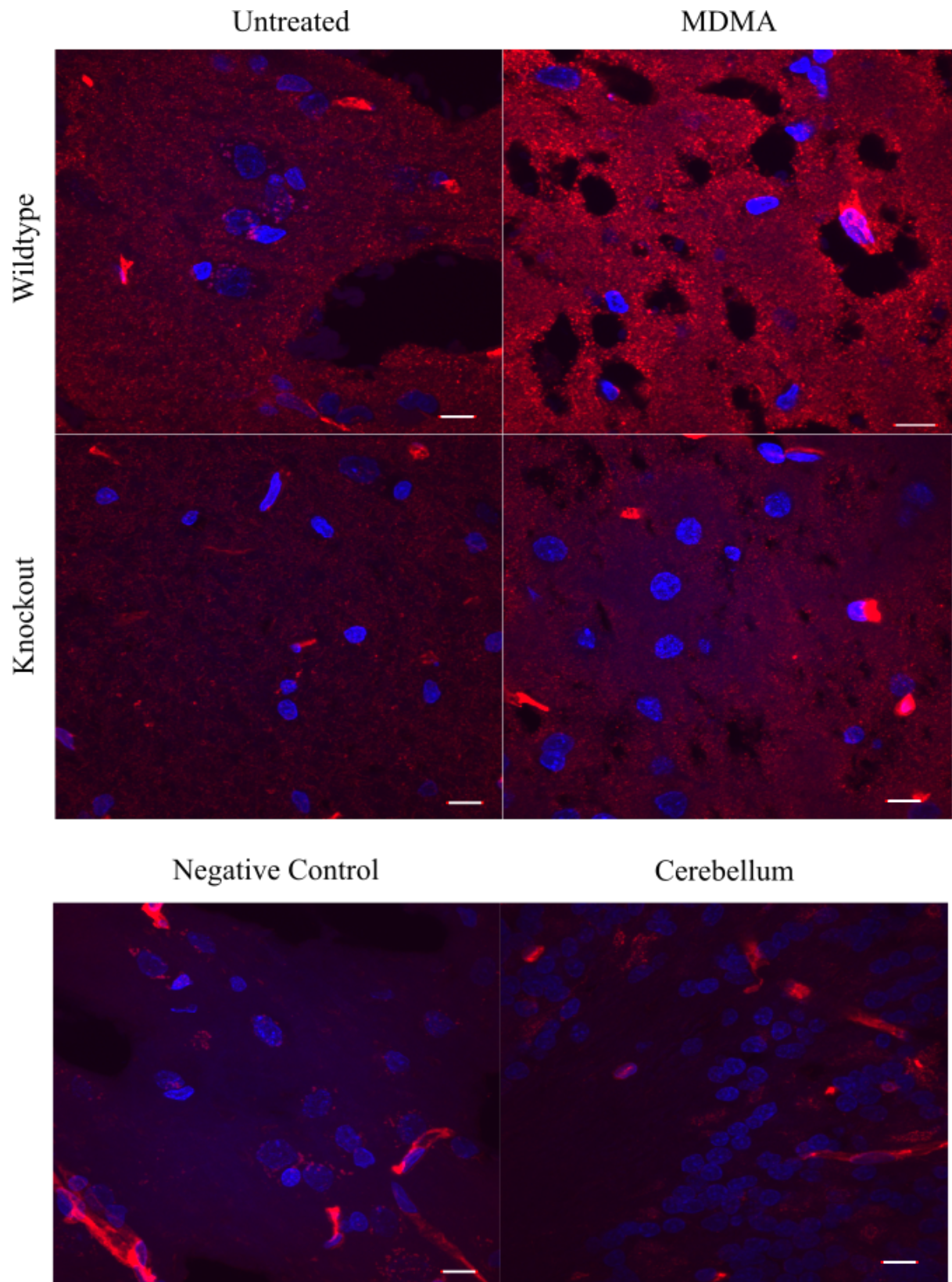


Figure 8. Drebrin expression in the frontal cortex of SERT KO and WT rats exposed to MDMA or left untreated (red) with DAPI (blue). Differences in fluorescent labelling are especially clear in the MDMA-treated WT frontal cortex. Exposure for all images was artificially increased by 50%. Scale bar = 5 μ m.

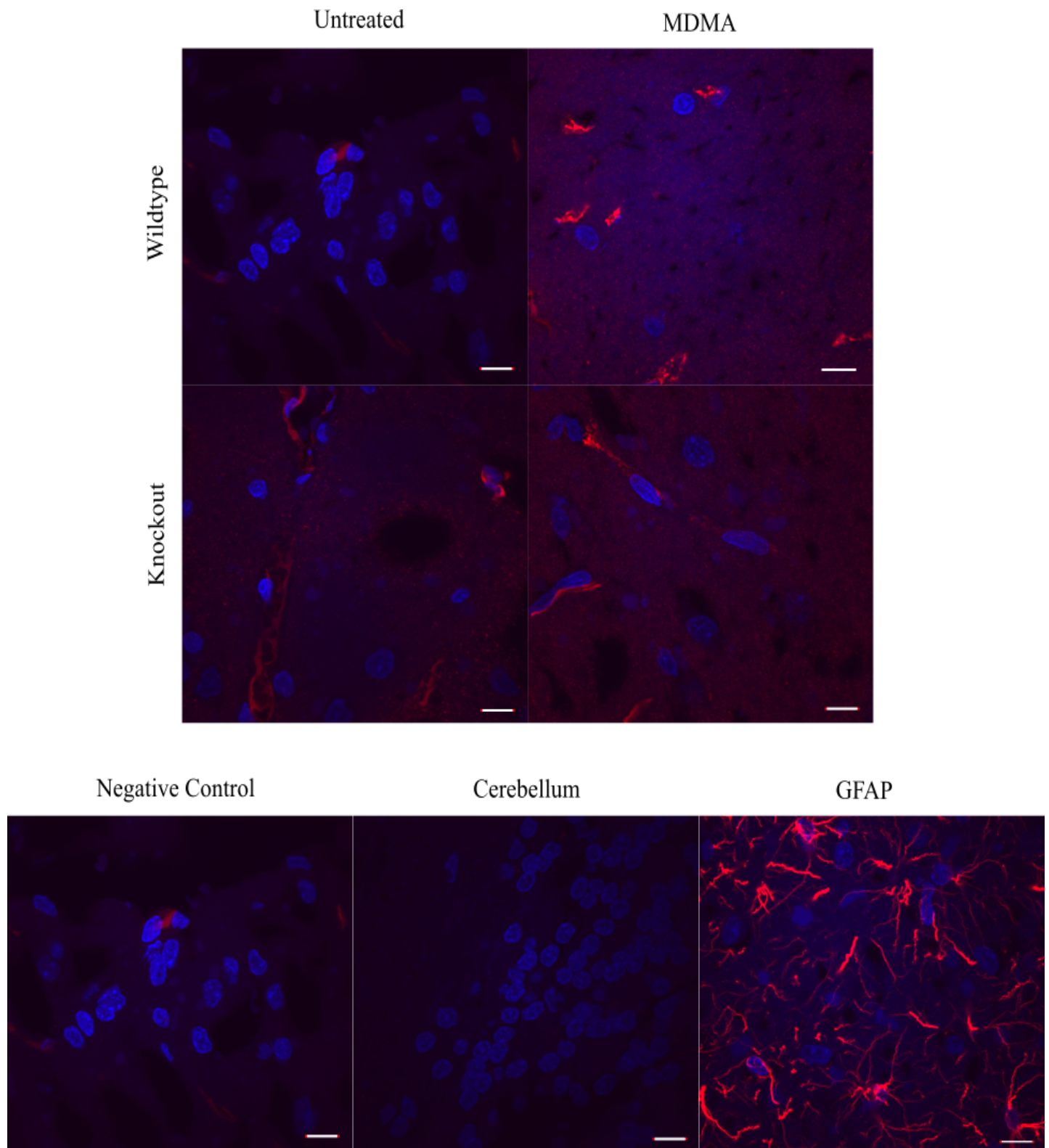


Figure 9. The second immunostaining test comparing drebrin expression in the frontal cortex of SERT KOs and WTs exposed to MDMA or left untreated (red) with DAPI (blue), alongside positive (GFAP; localised with Alexa Fluor 555) and negative controls. Results are qualitatively different from the first IF test comparing the four conditions. An absence of drebrin expression in the cerebellum indicates the protocol was successful. Exposure for all images was artificially increased by 50%. Scale bar = 10 μ m.

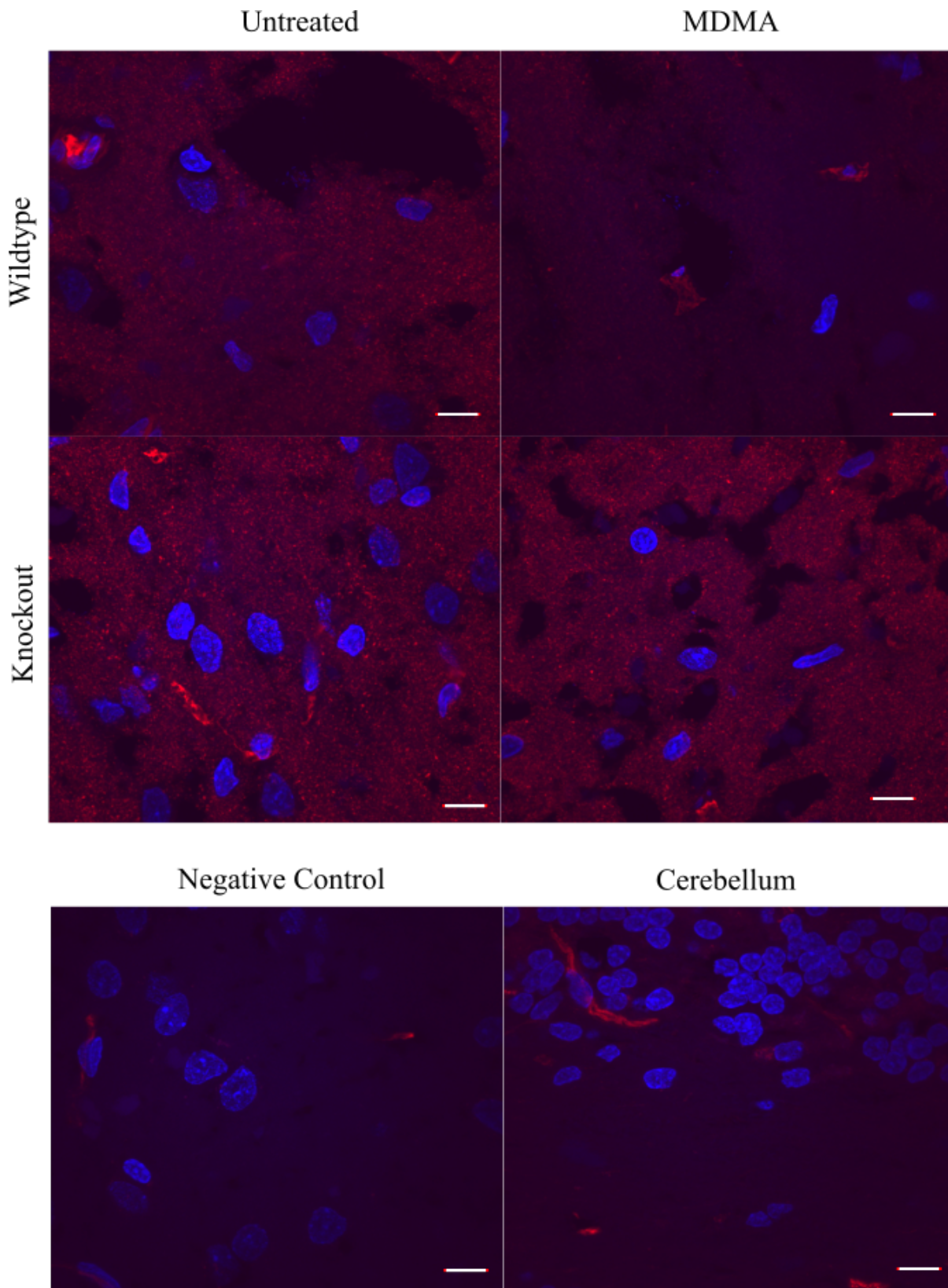


Figure 10. The third immunostaining test comparing drebrin expression in the frontal cortex of SERT KOs and WTs exposed to MDMA or left untreated (red) with DAPI (blue). An absence of drebrin expression in the cerebellum here and in Fig. 9 indicates the protocol was yet again successful. Exposure for all images was artificially increased by 50%. Scale bar = 10 μ m.

Discussion

The aim of this study was to optimise a protocol to examine neuronal drebrin expression and to characterise this expression in SERT KOs and WTs exposed to MDMA self-administration. It was hypothesised that inhibition of the SERT via genetic deletion and MDMA exposure would upregulate drebrin expression. After trialling protocols using enzyme-conjugated IHC and IF, a protocol was successfully optimised with the latter method. While an attempt was made to assess drebrin expression in MDMA-treated and MDMA-untreated SERT KOs and WTs, this proved unsuccessful using the current methodology. Qualitatively, patterns of drebrin expression were inconsistent within each condition. Therefore, no accurate conclusions can be drawn on how drebrin is differentially expressed as a function of SERT genotype and MDMA exposure.

One of the aims of this study was to optimise a protocol to detect drebrin in dendritic spines and ensure drebrin was binding to the appropriate target. Drebrin expression data are consistent with literature detailing drebrin expression in adult rodents. Fluorescence was punctate and characterised as small bulbous protrusions corresponding to dendritic spines (e.g. Figures 6-7). These features are consistent with previous studies using immunocytochemistry with fluorescence microscopy to detect drebrin *in vitro* (Koganezawa et al., 2017; Takahashi et al., 2009; Hanamura et al., 2018), as shown in Figure 11, as well as with diolistic labelling with IF (Jung et al., 2015).

Further evidence supporting the accurate binding of the antibody and optimisation of the protocol was seen when examining cerebellar drebrin expression. Across the immunostaining sessions, drebrin was not expressed in the cerebellum. Previous studies using enzyme-conjugated IHC show low cerebellar drebrin expression in the adult rat brain and high expression in the cortex (Hayashi et al., 1996). In the current study, fluorescence was abundant

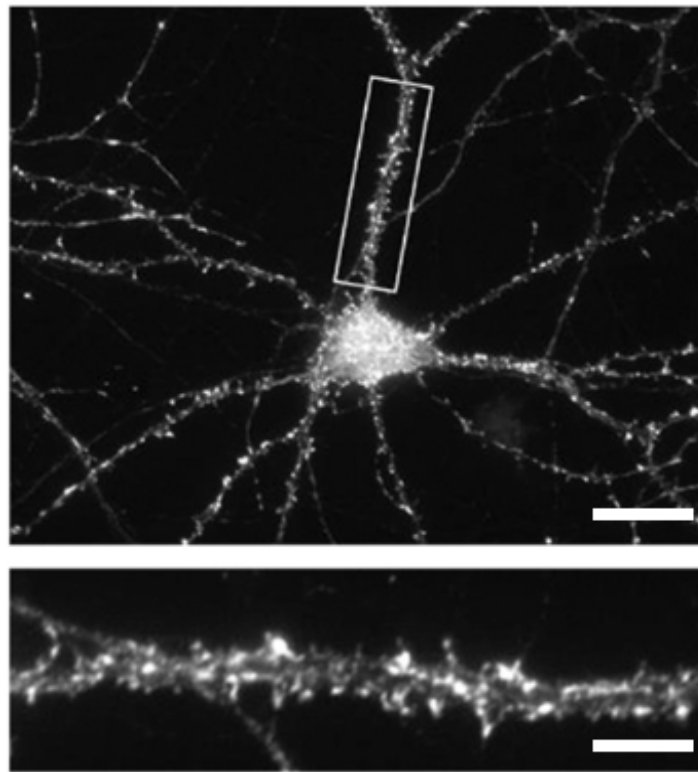


Figure 11. Cultured hippocampal neurons fixed at DIV 21, followed by immunocytochemical labelling of drebrin in dendritic spines. The pattern of drebrin expression in the current study resembles the drebrin localisation in dendritic spines shown here. Bottom panel is a zoomed-up image of the area contained within the rectangle in the top panel. Scale bars = 20 μ m (top panel), 10 μ m (bottom panel). Figure reproduced under Creative Commons Attribution-NonCommercial-No Derivatives 4.0 International License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>) from Koganezawa et al. (2017). These images are a cropped version of the original figure with scale bars added.

in the cortex but absent in the cerebellum, which is consistent with these previous findings. Although the drebrin antibody used in this study does not differentiate between drebrin E and drebrin A, the former isoform is not present in adult brain tissue (Hayashi et al., 1996; Hayashi et al., 1998). Further corroboration that the fluorescence in drebrin-labelled tissue indeed depicted drebrin A in dendritic spines could have been accomplished by co-labelling with postsynaptic markers such as PSD-95 (Hanamura et al., 2018). Nonetheless, the data presented in this study suggests that the protocol was optimised successfully.

Examining drebrin expression across MDMA-treated and MDMA-untreated SERT KOs and WTs proved to be more challenging for several reasons. Fluorescent labelling was ubiquitous in the frontal cortex even at high magnifications using confocal microscopy. The

inconsistency in patterns of drebrin expression across the biological repeats may indicate that subtler changes in expression are occurring that are not detectable using confocal microscopy. Therefore, more sensitive approaches are required to quantify drebrin expression in the future. Indeed, the most common method for quantifying drebrin expression is western blot analysis. Western immunoblotting is a common drebrin-detection method both *in vivo* and *in vitro* (Hayashi et al., 1998) and has been used to quantify the relative expression of drebrin E and A as a function of rat brain maturation (Aoki et al., 2005; Hayashi et al., 1996), as well as drebrin A expression after antisense knockdown of this isoform (Kobayashi et al., 2004) or in DAKO mice (Kojima et al., 2010). It has been used to detect drebrin using the M2F6 antibody employed in the current study, and a drebrin A-specific antibody, in hippocampal mice tissue (Kojima et al., 2016). Finally, western immunoblotting has also been used to quantify drebrin in human brain tissue from individuals with mild cognitive impairment (Counts et al., 2012). It is therefore apparent that the most efficacious method for drebrin quantification is western blot analysis, because this provides a more sensitive measurement of drebrin expression not evident with other techniques such as the IF methodology on frozen rat brain sections used in the current study.

Therefore, while dendritic spines may have been identified using confocal microscopy, a major limitation was that it was difficult to examine the subtle changes occurring in dendritic spine morphology as a function of drebrin expression in these treatment x genotype groups. Indeed, fluorescence and confocal microscopy techniques are not able to produce high resolution images of dendritic spines for quantitative analysis, due to refraction limitations in light microscopy (Schouten et al., 2014). In general, studies relying on fluorescent techniques alone to examine drebrin expression are scarce, unless it is used *in vitro* with immunocytochemistry so that dendritic spine clusters can be counted (Sekino et al., 2006; Takahashi et al., 2009; Hanamura et al., 2018). However, studies incorporating fluorescence

methods also use supplementary techniques that increase the efficacy of IF for examining drebrin expression. Jung et al. (2015) used diolistic labelling with DiI to visualise hippocampal dendritic spines as this allows for better assessment of dendritic spine morphology (Staffend & Meisel, 2011), and can be coupled with drebrin immunofluorescence. IF has also been used with super resolution microscopy to visualise drebrin A in dendritic spines in cultured hippocampal neurons, such as stochastic optimal reconstruction microscopy as demonstrated in Figure 12 (STORM; Koganezawa et al., 2017). Inherent limitations in fluorescence and confocal microscopy methods used in this thesis could have been overcome by employing techniques such as STORM or diolistic labelling to supplement the IF protocol and better visualise drebrin in dendritic spines.

Thus, it is clear that microscopy was another limitation in the methodology used here. Because confocal microscopy employed in the current study lacked super resolution features,

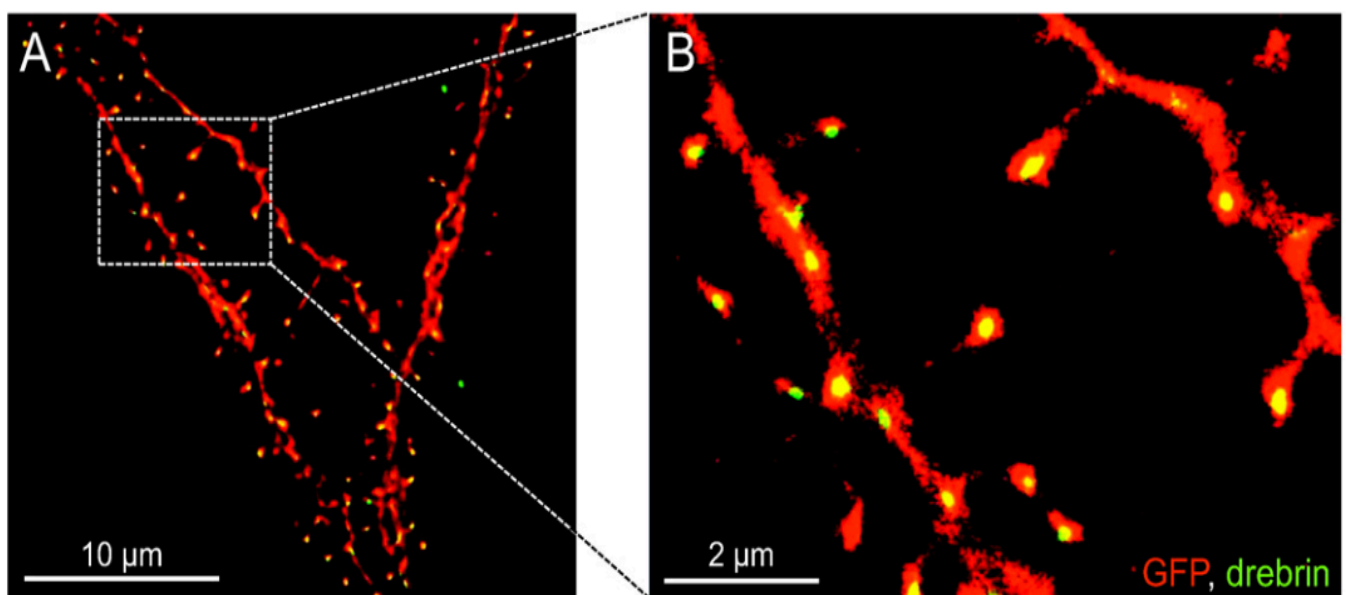


Figure 12. Drebrin A localisation in dendritic spines of cultured hippocampal neurons using super resolution microscopy (stochastic optimal reconstruction microscopy; STORM). Panel B is a zoomed-up image of the area contained within the white rectangle in Panel A. *Note.* GFP = Green fluorescent protein. Reproduced under Creative Commons Attribution-NonCommercial-No Derivatives 4.0 International License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>) from Koganezawa et al. (2017).

other microscopic techniques could also be used. As stated, this could include STORM (Koganezawa et al., 2017), but also electron microscopy. Aoki et al. (2005) used electron microscopy to discern expression of drebrin A in asymmetric and symmetric synapses on dendritic shafts and spines in the adult rat cortex and hippocampus. Not only did this allow for observation of pre- and post- synaptic ultrastructure, it also allowed for drebrin quantification at the level of shafts or spines. This differentiation of drebrin-labelling between shafts and spines was not feasible using confocal microscopy in the present study. Quantification using electron microscopy showed that drebrin was absent presynaptically and was only found in asymmetric synapses. Therefore, alternative methods to confocal microscopy such as electron microscopy would have been more suitable for the aims of this study because of its ability to quantify this protein at ultrastructure levels.

While a protocol to identify drebrin in brain tissue using IF was optimised, the question remains whether drebrin expression is increased in SERT KOs or after MDMA exposure. As stated, future studies should employ more sensitive methods to delineate how drebrin expression is altered after genetic deletion of the SERT and MDMA self-administration. Indeed, an effective approach would be to apply multiple methods simultaneously. Previous studies have used both histological and western immunoblotting techniques together (Song et al., 2008; Hayashi et al., 1996), which could also be employed here. This combination would allow for more effective comparisons of drebrin expression across MDMA-treated and MDMA-untreated SERT KOs and WTs, such that expression patterns obtained from histological data can be supported with western immunoblotting data. Moreover, if higher resolution microscopy methods such as STORM or electron microscopy are also employed, drebrin quantification on brain sections could be possible through drebrin cluster sampling (Aoki et al., 2005). Finally, primary neuronal cultures are another effective method. The effect of SERT inhibition (via MDMA exposure) on drebrin expression could be examined in primary

neuronal cultures. This would be similar to how Roppongi et al. (2013) quantified drebrin in dendritic shafts and spines after exposure to a 5-HT_{2A} agonist in cultured hippocampal neurons. Utilising these various methods would better assess the hypothesis that inhibition of the SERT via genetic deletion or MDMA exposure increases drebrin expression.

The current study aimed to optimise a method to examine drebrin expression in cortical rat brain tissue, and to determine whether drebrin is differentially expressed in SERT KOs and those exposed to MDMA self-administration. Both enzyme-conjugated IHC and IF were trialled, but the protocol was optimised using IF. However, IF only allowed for characterisation of drebrin expression at a qualitative level. Drebrin quantification was not feasible using the current methodology. Therefore, no supporting conclusions can be drawn on whether drebrin expression is altered after SERT inhibition. Future studies should take note of the limitations in the current study and employ more sensitive methods such as western immunoblotting, electron microscopy, and primary neuronal cultures. This would further determine the nature of the relationship between drebrin, drug exposure, and the SERT, which remains a novel and understudied area.

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